

CS 349 - Database and Information Systems

Project Report

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1 Goals of the Project

Our project, **WellVault**, a **Centralized Health Record System (CHRS)**, aims to provide a unified platform for managing and accessing patient health records, health insurance details, and medical data. This system will make healthcare more efficient by bringing together medical records, prescriptions, and insurance details from different hospitals and clinics into one secure and easy-to-access place.

One of the key motivations behind this system is emergency situations, where quick access to a patient's complete medical history, allergies, ongoing treatments, and insurance details can be life-saving. Doctors can make informed decisions instantly, while insurance details being readily available will streamline the billing and claims process during crucial times.

2 Implementations from User Perspective

2.1 Patient as the User

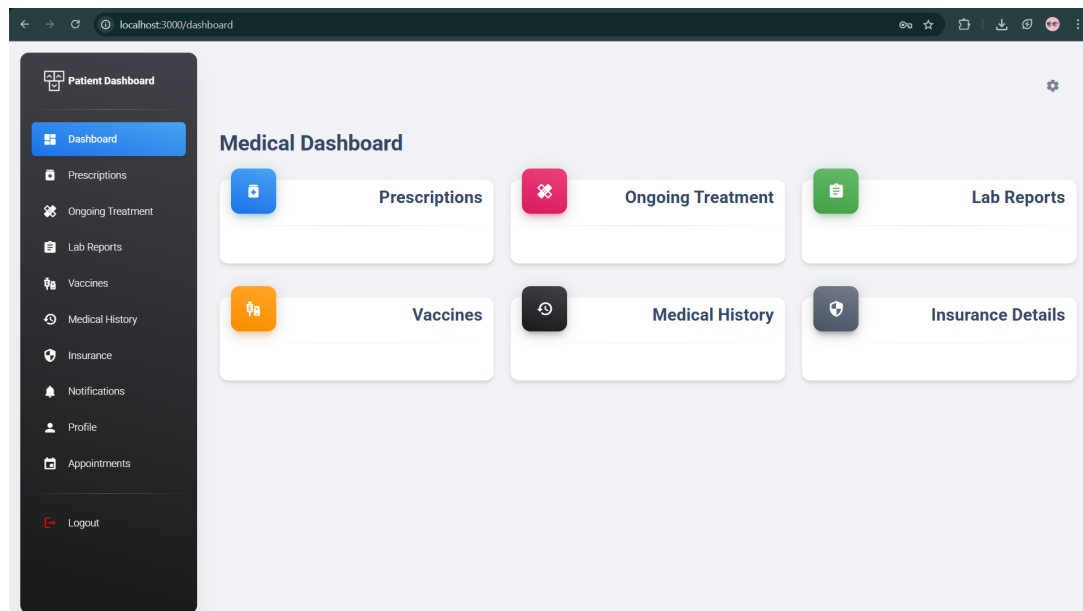


Figure 1: Patient Dashboard from the User Perspective

- **Patient Signup**

- Fields: Username, Password, Full Name, Date of Birth, Gender (dropdown), Mobile (10-digit), Emergency Contact ($\neq$ mobile), Address
- Validations: Unique username and emergency contact $\neq$ mobile
- Mandatory: Terms and Conditions agreement checkbox + Sign Up button

- **Patient Login**

- Inputs: Username + Password
- Sign In Button

- **Prescriptions**

- Clean grid layout showing all medications
- Big “+” button to add new prescriptions
- Edit/delete icons on every medication card

- Pop-up form with fields medicine name, dosage (e.g., “500mg”), start and end date pickers, and doctor’s name
- **Ongoing Treatments**
 - Clean grid layout showing all treatments
 - Big “+” button to add new treatments
 - Edit/delete icons on every treatment card
 - Pop-up form with fields for treatment name, doctor in charge, start and end date pickers, status dropdown (Ongoing, Paused, Completed), and description box
- **Lab Reports**
 - Card-based display of all lab reports
 - “+” button to add new reports
 - Edit/delete options for each report
 - Upload form with description field, file upload for PDF/image reports
 - Direct download links for saved reports
- **Vaccines**
 - Card grid showing vaccination history
 - “+” button to add new vaccine records
 - Edit/delete options for each entry
 - Form with dropdown menu of common vaccines, dose number input, year administered field, hospital/clinic name
- **Medical History**
 - Timeline view of medical records in card format
 - “+” button to add new health events
 - Edit/delete options for each record
 - Entry form with date picker (past dates only), hospital/clinic name field, diagnosis description
- **Insurance**
 - Policy search box with number/provider fields + verify button
 - Verified policy display (coverage dates, claim limits)
 - Two policy addition methods:
 - * External: Form with dates/description/amount
 - * Pre-approved: Card grid with plans/duration
- **Notifications**
 - Upcoming appointment cards (doctor, date/time, hospital)
 - “No appointments” placeholder message
 - Health resource links: Recipes • News • Guides • Self-tests
 - × button to remove alerts
- **Profile**
 - Health Stats: Height/weight • Blood group • Allergies • Treatment status
 - Avatar: 10+ cartoon options • Upload custom photo • Preview before saving
 - Editable Fields:
 - * Single form for personal/health info
 - * Blood type dropdown
- **Appointments**
 - Booked appointments shown in clean card layout:
 - * Hospital name & doctor details

- * Appointment date (DD/MM/YYYY format)
- * Time slot in 12-hour format (e.g., 02:30 PM)
- * Payment status indicator
- Empty state message: “No appointments yet” when no bookings exist
- Automatic updates after new bookings

2.2 Hospital as the User

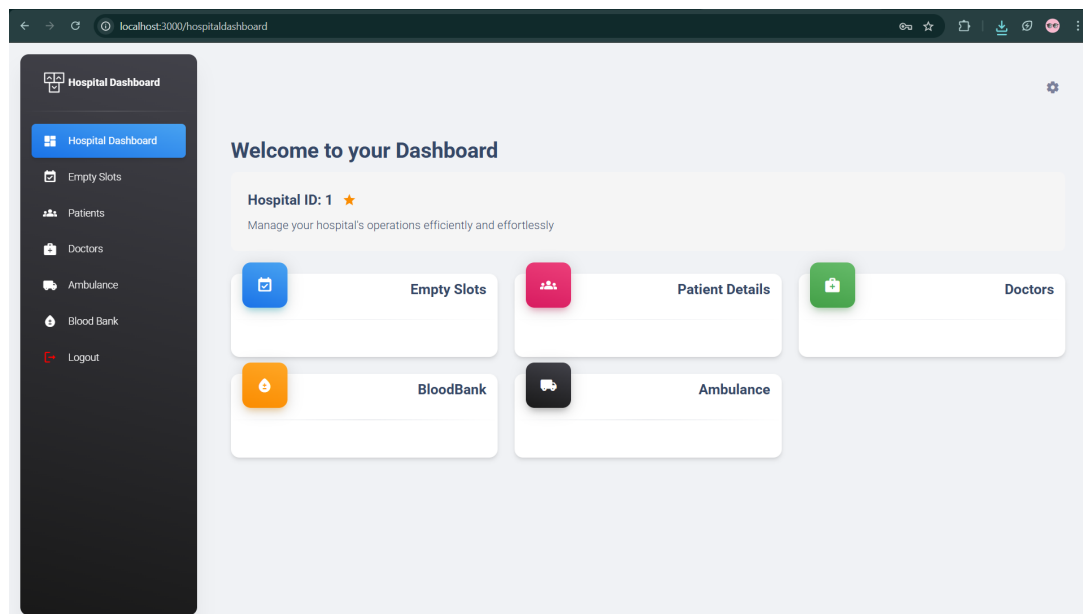


Figure 2: Hospital Dashboard from the User Perspective

- **Signup**
 - Fields: Hospital Name, Pincode (6-digit), Address
 - Dropdowns: Hospital Type (Public/Private), Ambulance/Blood Bank Availability (Yes/No)
 - Mandatory “Agree to Terms” checkbox + Sign Up button
- **Signin**
 - Inputs: Hospital ID (auto-generated), Password
 - Sign In button
- **Empty Slots**
 - Date selection input (default: current date)
 - Grid layout showing doctor cards
 - Each doctor card displays:
 - * Doctor’s name
 - * List of available time slots (e.g., “10:00am- 11:00am”)
 - * “No available slots” if fully booked
 - “No doctors available for the selected date” message if no data
- **Patients**
 - Three search filters at the top: Name input (text field), Doctor name input (text field), Date picker (calendar dropdown)

- Grid of patient cards displaying: Patient name, Doctor assigned, Appointment date (YYYY-MM-DD format), Appointment time slot
- Dynamic filtering updates results as users type/select filters
- “No appointments scheduled till now” message if no matching data
- **Doctors**
 - Two search filters at the top: Doctor name input (text field), Speciality filter (text field)
 - Grid of doctor cards showing: Doctor’s name (bold text), Speciality field (e.g., “Cardiology”)
 - Real-time filtering updates as users type
 - “No doctors found” message if no matches
- **Ambulance**
 - Add Ambulance Form: Vehicle number input (validates format XX 99 XX 9999), Type dropdown (Emergency/Non-Emergency/ICU/Transport), and then click “Add Ambulance”
 - Each Ambulance Card displays: Vehicle number, Ambulance type, Availability toggle (checkbox labeled Available/Unavailable), Red “Delete” button with hover effects
- **Blood Bank**
 - Add Blood Bank Form: Blood type dropdown (A+/A-/B+/B-), Count input (number field, disabled until blood type selected - can’t), and then click “Add Blood Bank” button
 - Each Blood Bank card shows: Blood type, Adjustable count with +/- buttons, and a Red delete button with hover effects

3 Architecture of our Solution

The system follows a **three-tier architecture**, consisting of:

- **Frontend (Client Tier)**
 - Patients, hospital staff (admins, doctors) use it
 - Allows users to view, create, update, and delete data like:
 - * Patient prescriptions
 - * Appointments
 - * Ambulance requests/status
 - * Hospital slots/schedules
- **Backend (Application Tier)**
 - Technology stack: Node.js
 - Responsibilities:
 - * Handle HTTP requests from the frontend
 - * Enforce authentication and authorization
 - * Communicate with the database
- **Database (Data Tier)**
 - Technology: Relational (PostgreSQL, MySQL)
 - Holds data about:
 - * Patients
 - * Doctors
 - * Hospitals

- * Prescriptions, treatments
- * Appointment slots, ambulance details

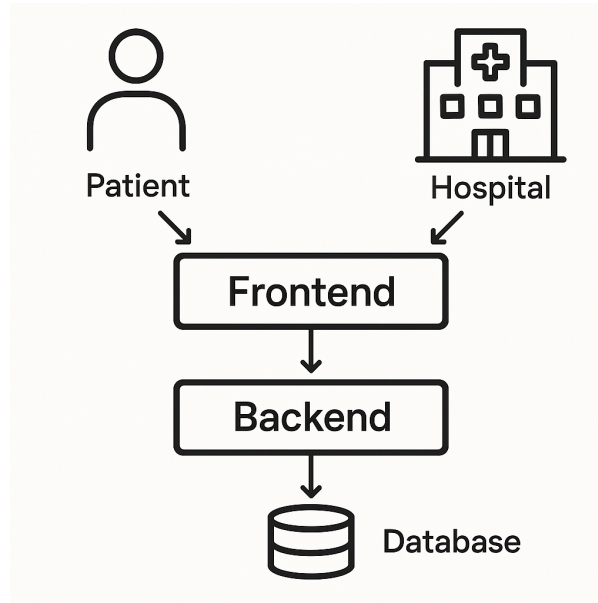


Figure 3: Architecture

4 Frontend Functionality

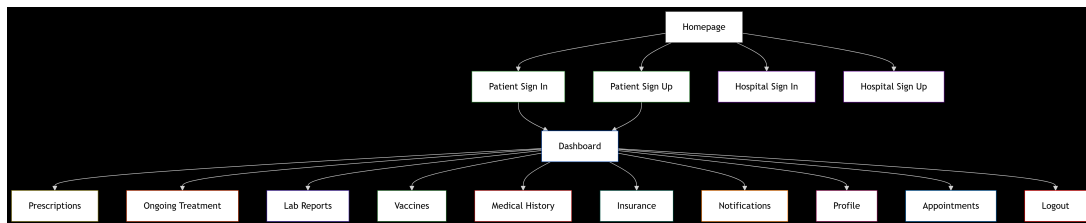


Figure 4: Patient Dashboard Flowchart

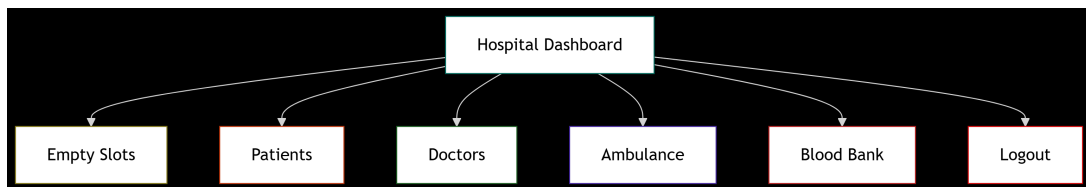


Figure 5: Hospital Dashboard Flowchart

5 Backend Functionality

5.1 Authentication and Authorization

- Session Management for users (username-based) and hospitals (hospital ID-based)
- Secure password handling with bcrypt hashing (10 rounds)

- Middleware (isAuthenticated, is Hosp Authenticated) for route protection
- CORS configuration for React frontend integration

5.2 Patient Profile Management

5.2.1 Personal Data

- Stores basic details like name, date of birth, gender, and address
- Saves emergency contacts and mobile numbers (10-digit format enforced)
- Prevents duplicate mobile numbers/username registrations
- Allows users to edit/update information securely (in Profile section)

5.2.2 Medical Records

- Tracks vaccination history with dose numbers and hospital details
- Stores lab reports as PDF files (upload/download supported)
- Maintains blood group records and allergy information
- Shows complete medical history with diagnosis dates

5.2.3 Treatment Tracking

- Records current medications with dosage instructions
- Tracks treatment schedules (start/end dates)
- Flags if the medication end date is before the start date
- Shows active/completed treatments with doctor details

5.2.4 Insurance Integration

- Links insurance policies to user accounts
- Auto-generates unique policy numbers
- Shows coverage details and claim limits
- Displays policy validity period (start/end dates)
- Allows users to add their existing insurance plans
- Lets users browse/avail pre-approved health plans

5.2.5 Appointment System

- Shows doctor availability time slots
- Books appointments only for future dates
- Marks time slots as booked after reservation
- Tracks appointment history with hospital details
- Integrates payment system (500 rupees fee per appointment)

5.3 Hospital Management Portal

- Updates doctor schedules and appointment slots based on the date
- Views patient appointment lists with timings
- Stores doctor details
- Manages ambulance availability (shows vehicle type/numbers)
- Tracks blood bank stock by blood group
- Toggles ambulance/blood bank availability status

6 Implementation including usage of tools such as AI code generation

Our development process used AI-assisted tools (ChatGPT-4 and DeepSeek). Key implementations include:

- **Dummy Data Generation**
- **Frontend Enhancement:**
 - Material-UI components with accessibility improvements
 - Responsive design suggestions
 - Dynamic form validation rules for patient registration flows

7 Future Work

- **Doctor Portal:** Introduce a dedicated portal for doctors where they can manage patient prescriptions, view medical histories, and add new treatment records. All updates made by doctors would automatically reflect on the patient’s portal, ensuring seamless and real-time medical record synchronization.
- **Integrated Payments Interface:** Develop a secure, fully functional payments module that allows patients to pay for appointments, treatments, and other services directly through the platform. This can include support for various payment modes like UPI, credit/debit cards, and mobile wallets.
- **Insurance Management Portal:** Build a system where insurance companies can register and upload new health insurance plans. Patients can browse and apply for suitable plans through an “Avail New Policies” section. Additionally, insurance providers can securely access relevant medical records and hospital bills to simplify the claim verification process.
- **Symptom-Based Doctor Recommendations:** Implement a system where patients can input their symptoms (via structured forms or natural language) to receive doctor recommendations.

8 Conclusion

WellVault successfully demonstrates a functional Centralized Health Record System that bridges patients and healthcare providers through secure, real-time data access. Built using React, Node.js, PostgreSQL, our web application streamlines medical record management, appointment scheduling, and insurance integration while adhering to data privacy standards. Future enhancements like a doctor portal and payment gateway (as outlined in Section 7) promise extended functionality. This project highlights the transformative potential of integrated digital systems in healthcare.

Git Repository: <https://github.com/PiyushiAnand/WellVault.git>